

Exercise 5: Implementing the Decorator Pattern

CODE:

```
// Component Interface
interface Notifier {
    void send(String message);
}

// Concrete Component
class EmailNotifier implements Notifier {
    public void send(String message) {
        System.out.println("Email Notification: " + message);
    }
}

// Abstract Decorator
abstract class NotifierDecorator implements Notifier {
    protected Notifier notifier;
    NotifierDecorator(Notifier notifier) {
        this.notifier = notifier;
    }
    public void send(String message) {
        notifier.send(message);
    }
}

// SMS Decorator
class SMSNotifierDecorator extends NotifierDecorator {
    SMSNotifierDecorator(Notifier notifier) {
        super(notifier);
    }
    public void send(String message) {
        super.send(message);
    }
}
```

```

        System.out.println("SMS Notification: " + message);
    }
}

// Slack Decorator
class SlackNotifierDecorator extends NotifierDecorator {

    SlackNotifierDecorator(Notifier notifier) {

        super(notifier);
    }

    public void send(String message) {

        super.send(message);

        System.out.println("Slack Notification: " + message);
    }
}

// Main Class
public class Main {

    public static void main(String[] args) {

        System.out.println("Email Notification");

        Notifier email = new EmailNotifier();

        email.send("Task Completed");

        System.out.println();

        System.out.println("Email + SMS Notification");

        Notifier emailSMS = new SMSNotifierDecorator(new EmailNotifier());

        emailSMS.send("Task Completed");

        System.out.println();

        System.out.println("Email + SMS + Slack Notification");

        Notifier allNotifications =

            new SlackNotifierDecorator(

                new SMSNotifierDecorator(

                    new EmailNotifier()));
    }
}

```

```
        allNotifications.send("Task Completed");  
    }  
}
```

OUTPUT:

```
Email Notification  
Email Notification: Task Completed  
  
Email + SMS Notification  
Email Notification: Task Completed  
SMS Notification: Task Completed  
  
Email + SMS + Slack Notification  
Email Notification: Task Completed  
SMS Notification: Task Completed  
Slack Notification: Task Completed
```